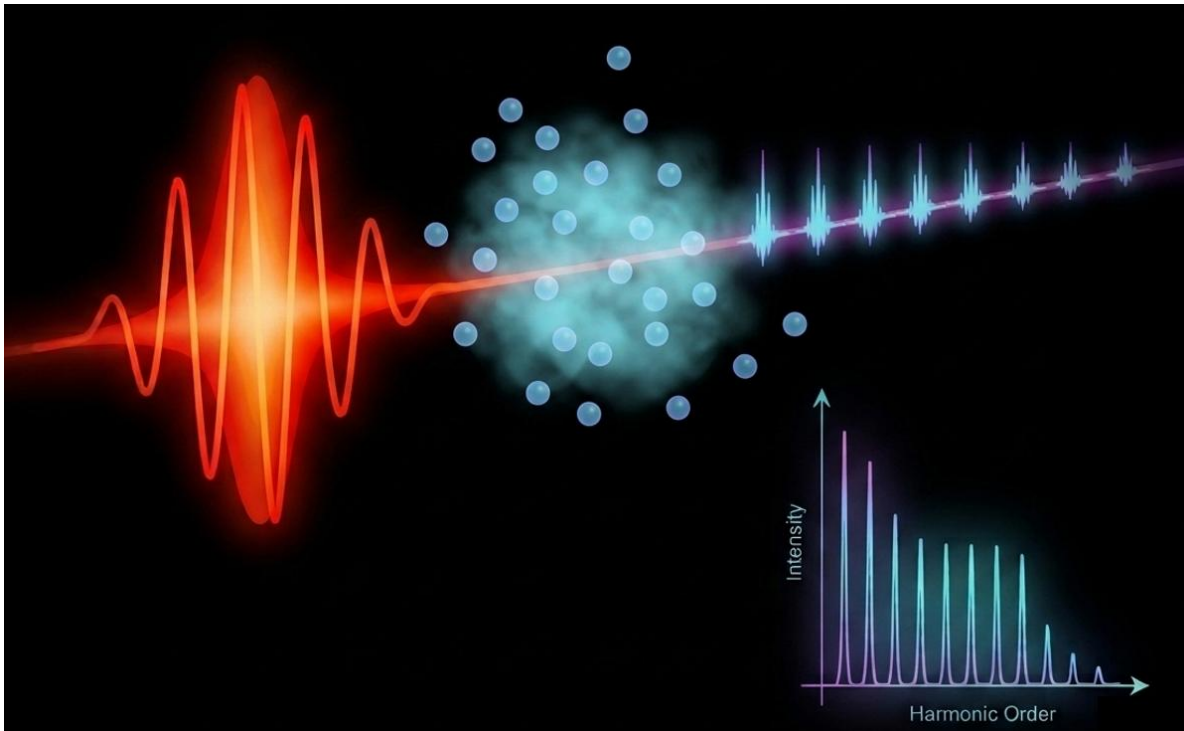


High harmonic generation and ultrafast electron dynamics simulations



IMPRS UFAST focus course

17th, 18th , 24th, 25th August
9:00-12:00h

Alba de las Heras
Theory department

UNTERSTÜTZT VON / SUPPORTED BY



Alexander von
HUMBOLDT
STIFTUNG



Outline

Session 1: Fundamentals of Strong-Field Physics

Session 2: Microscopic modeling of laser-driven electron dynamics and spectroscopic signatures in high harmonic generation

Session 3: Macroscopic picture of high harmonic generation and control of the properties of structured attosecond pulses

Session 4: Solid-state attosecond physics

Contents of Session 4:

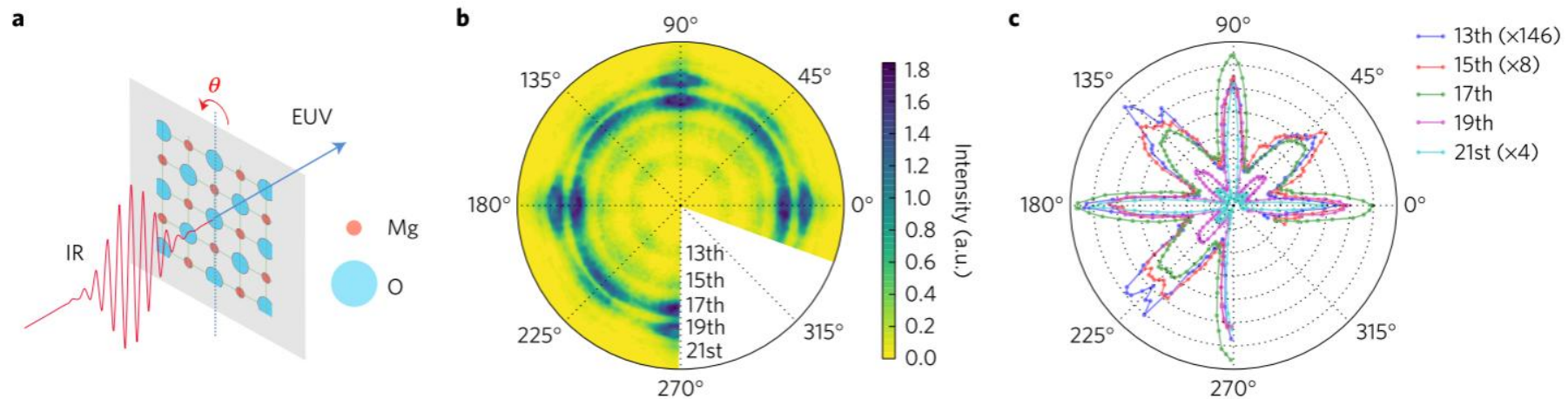
Fundamentals of high harmonic generation in solids

Illustrative research lines in solid-state high harmonic generation

Using Octopus code for TDDFT calculations in solids

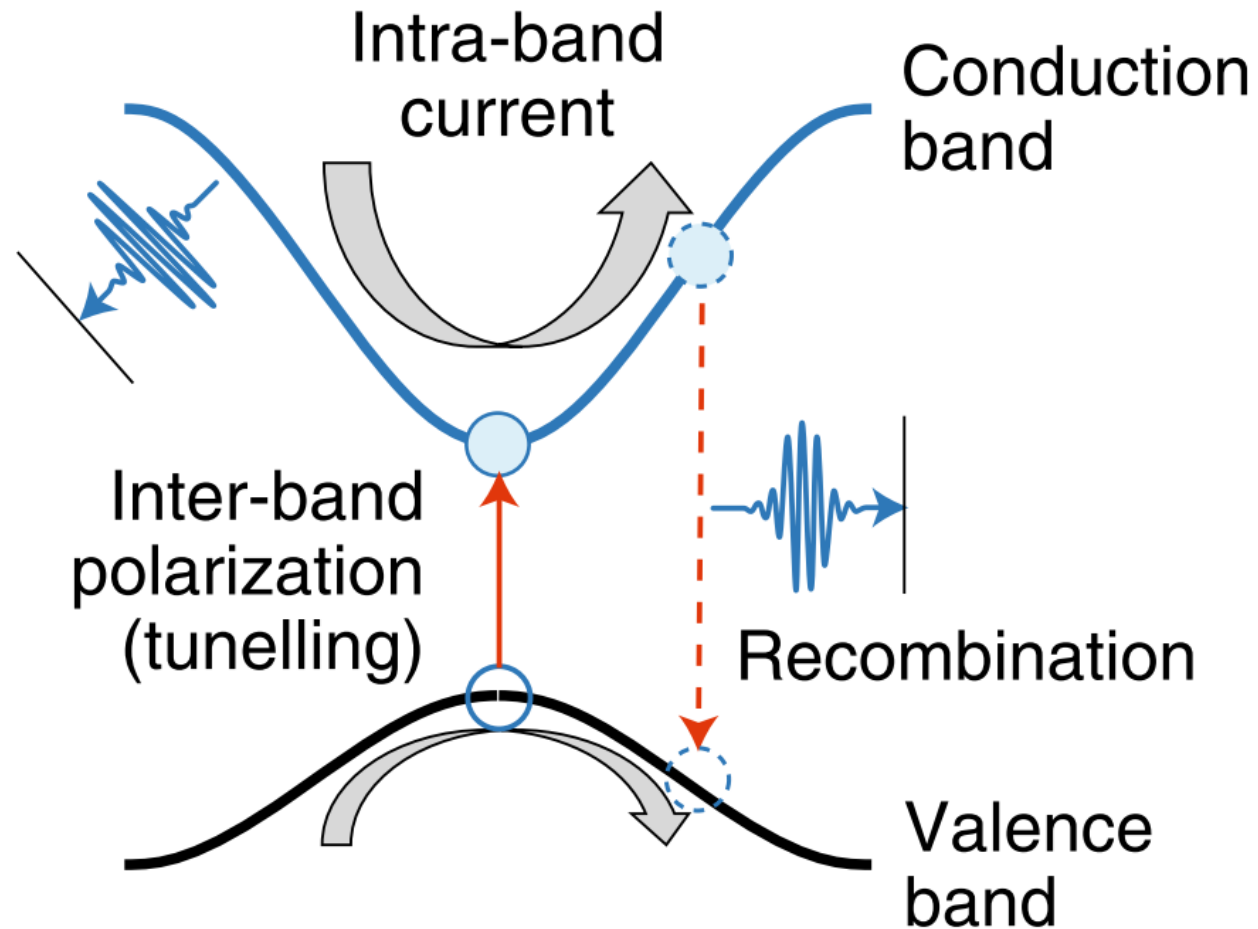
Attosecond pump-probe spectroscopy in solids

Fundamentals of high harmonic generation in solids



Anisotropic high-harmonic generation in bulk crystals.
Y. S. You, D. A. Reis & S. Ghimire. *Nat. Physics* (2017)

High-order harmonic generation (HHG) in solids



S. Ghimire & D. A. Reis, Nature Physics (2019)

First experiment on high harmonic generation in solids

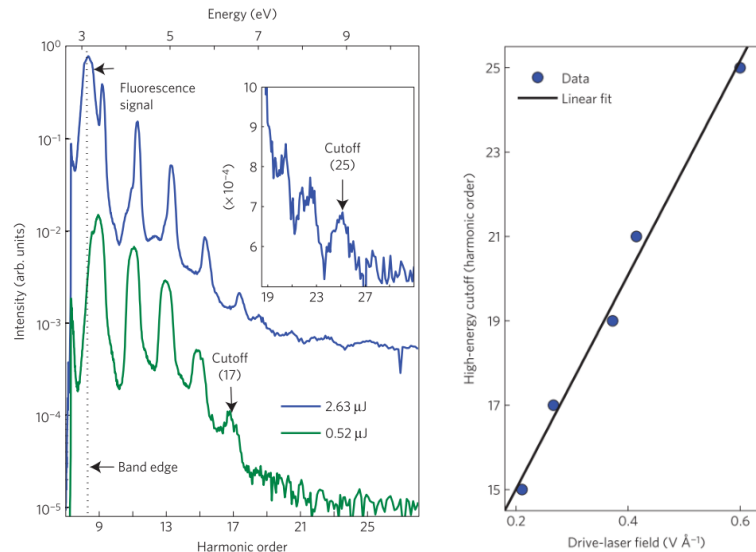
LETTERS

PUBLISHED ONLINE: 5 DECEMBER 2010 | DOI: 10.1038/NPHYS1847

nature
physics

Observation of high-order harmonic generation in a bulk crystal

Shambhu Ghimire¹, Anthony D. DiChiara², Emily Sistrunk², Pierre Agostini², Louis F. DiMauro² and David A. Reis^{1,3}★



The energy cutoff depends linearly on the driving electric field amplitude!

Differences in HHG from solids vs atoms

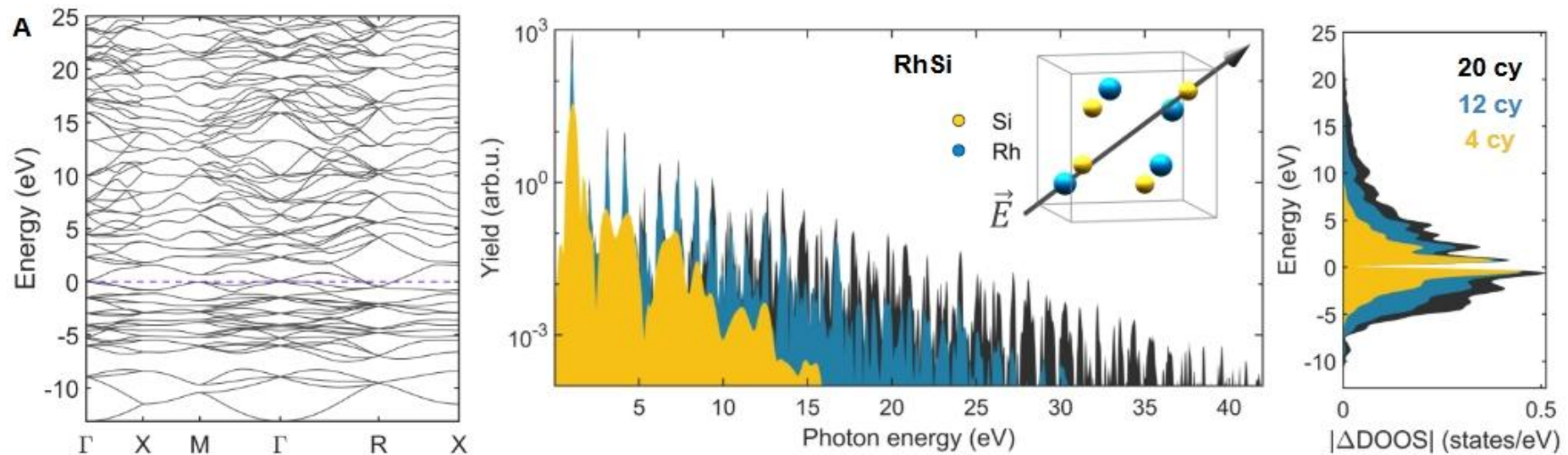
In solids, HHG is influenced by:

- Crystal symmetries and band structure
- Interband and intraband contributions
- Electron-hole interaction (e.g. exciton formation)

Spectroscopic signatures of solid-state HHG departing from the atomic picture:

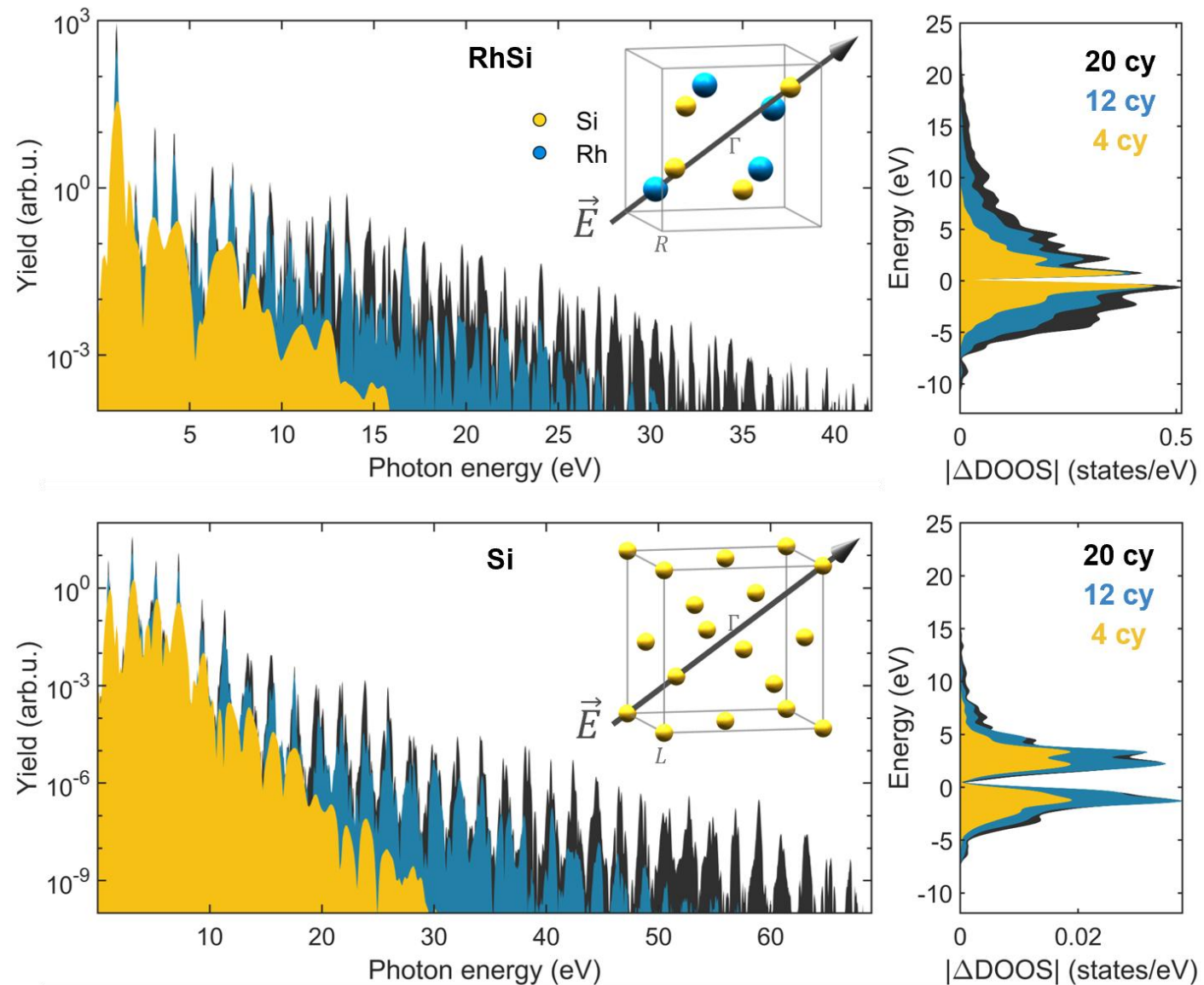
- Formation of multiple plateaus even in a single-electron framework
- The cutoff law of HHG in atoms is no longer valid. In solids, the energy cutoff is typically determined by the interband mechanism, so that the maximum photon energy is given by the maximum energy of electron-hole recombinations. It shows a linear dependence on the field strength. The frequency dependence influences the cutoff energy of the interband transition, but not of the intraband contribution. The cutoff can depend on the pulse duration.

Role of pulse duration and excited ladder electrons in high harmonic generation

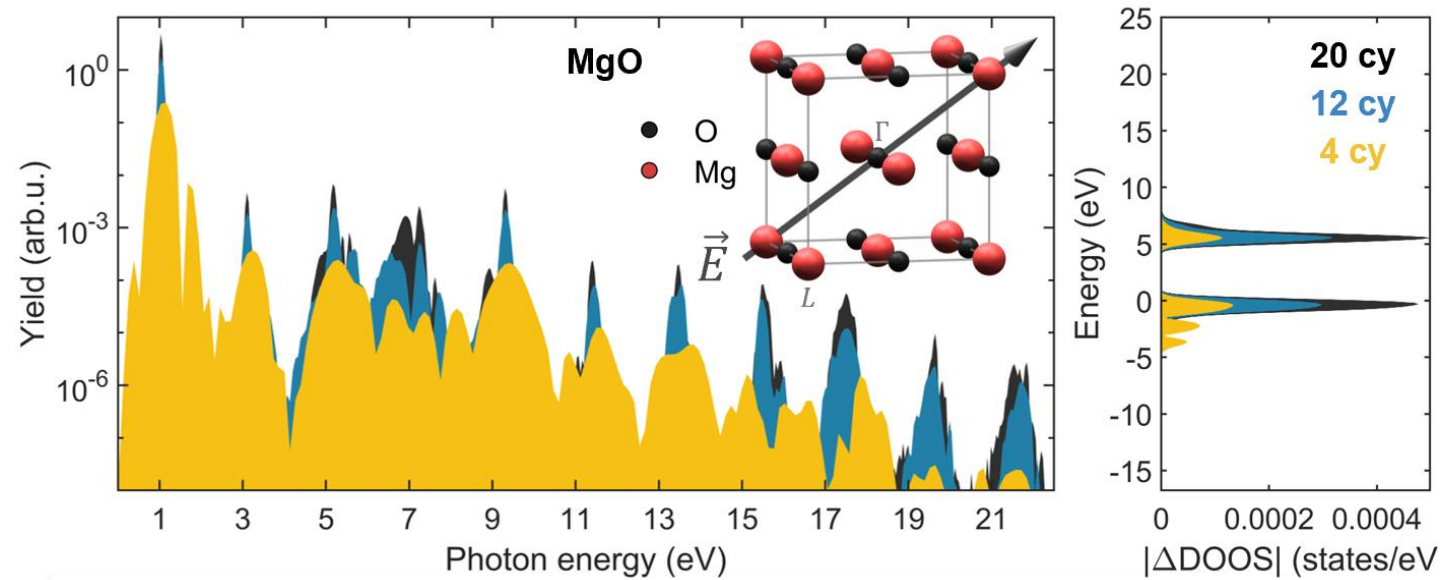


A. de las Heras *et al.* *Nano Letters* (2026)

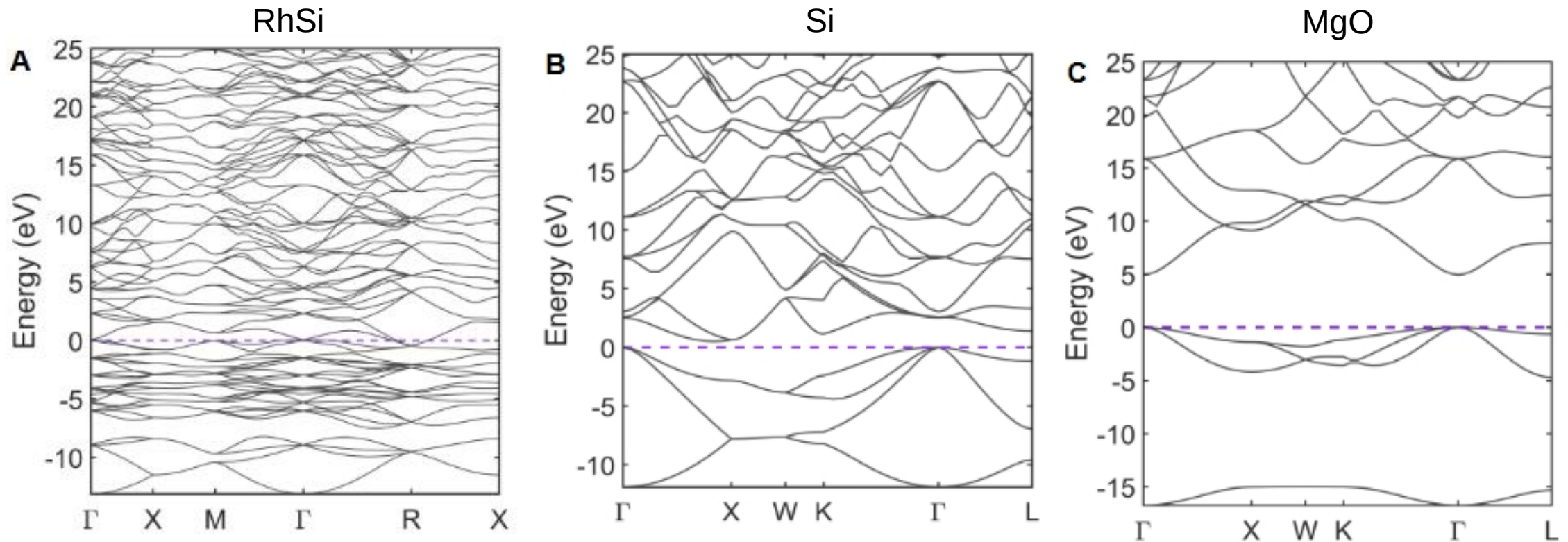
Pulse-duration sensitive high harmonic generation



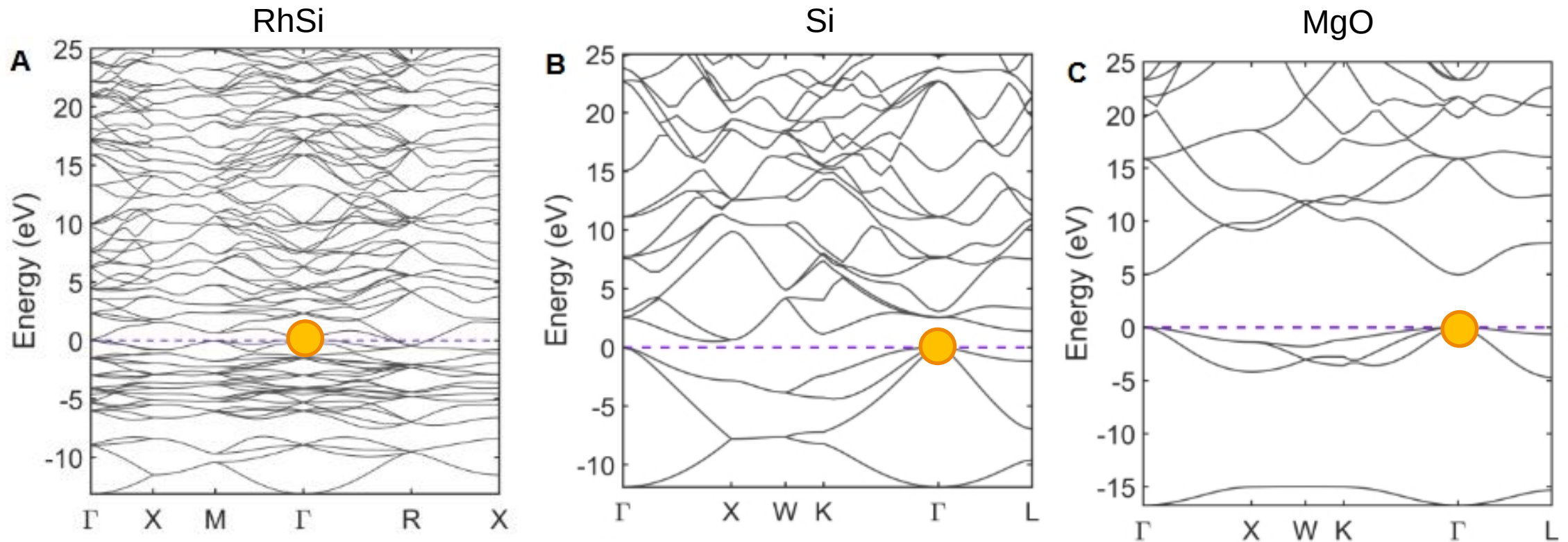
MgO does not show a substantial cutoff dependence on the driving pulse duration



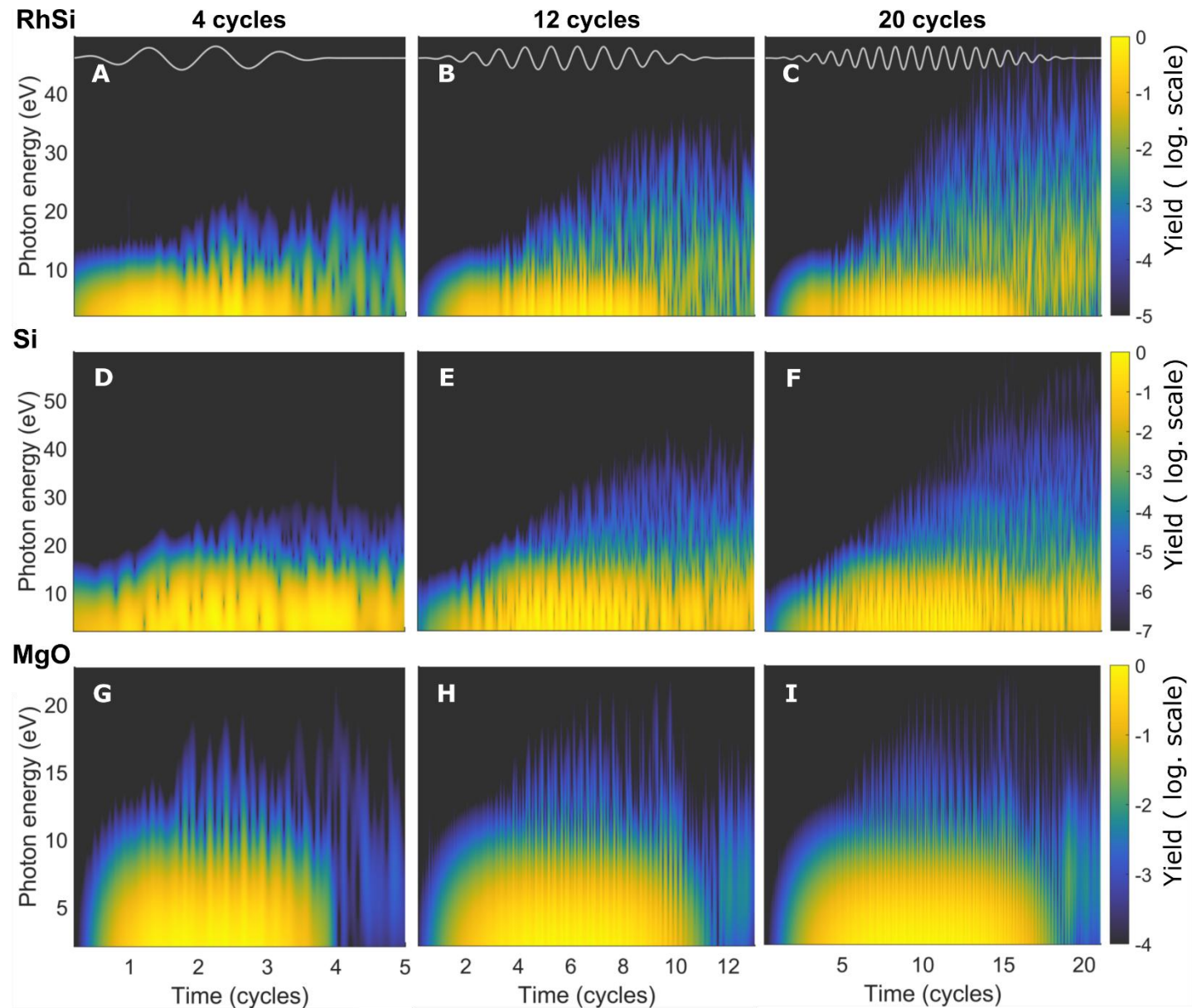
- Band crossings and narrow bandgaps increase the dipole coupling to higher energy conduction bands, thus favoring the build-up of an electron excitation ladder.
- The subsequent electron-hole recombination from a higher energy transition results in a higher photon energy emission.



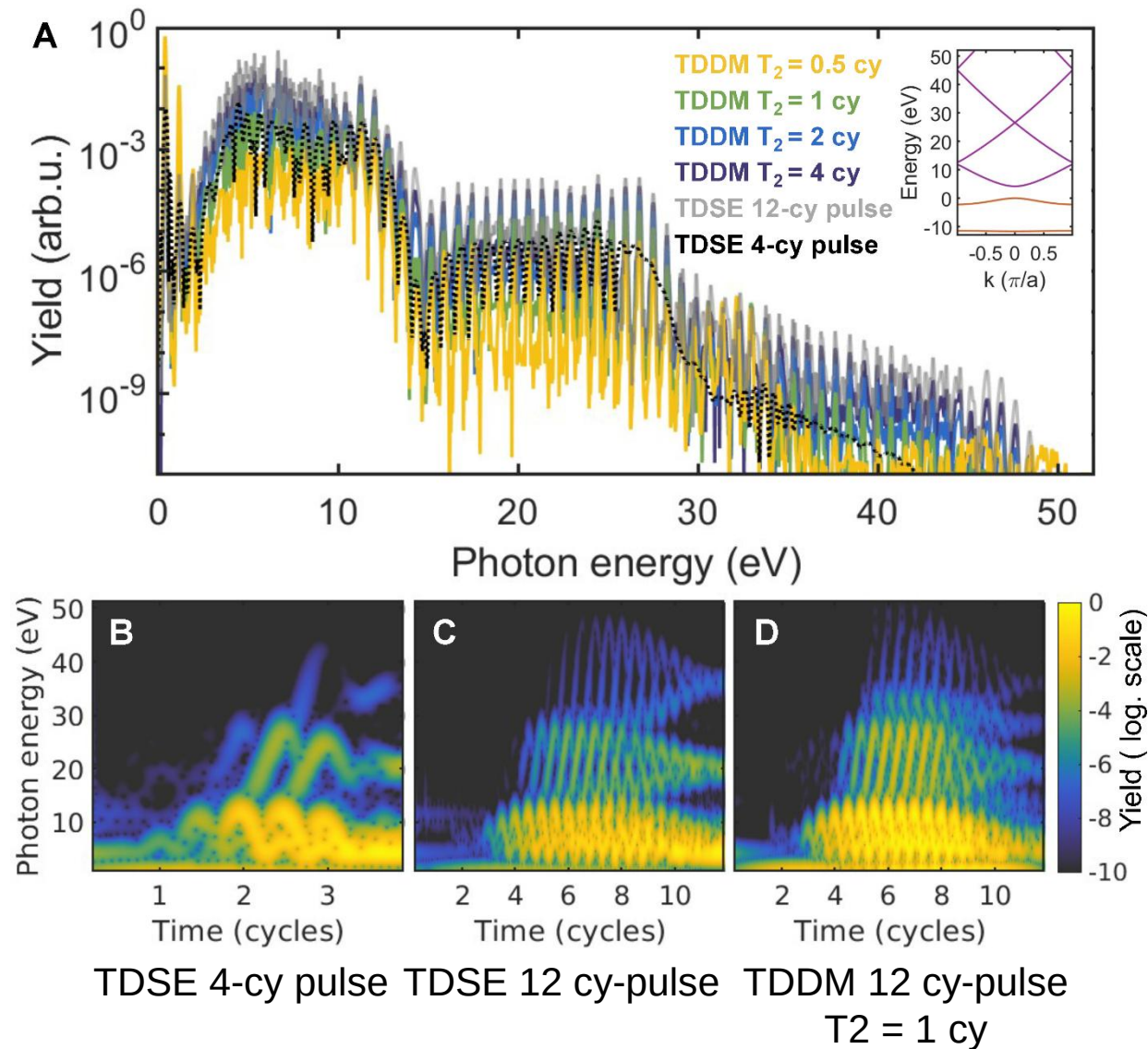
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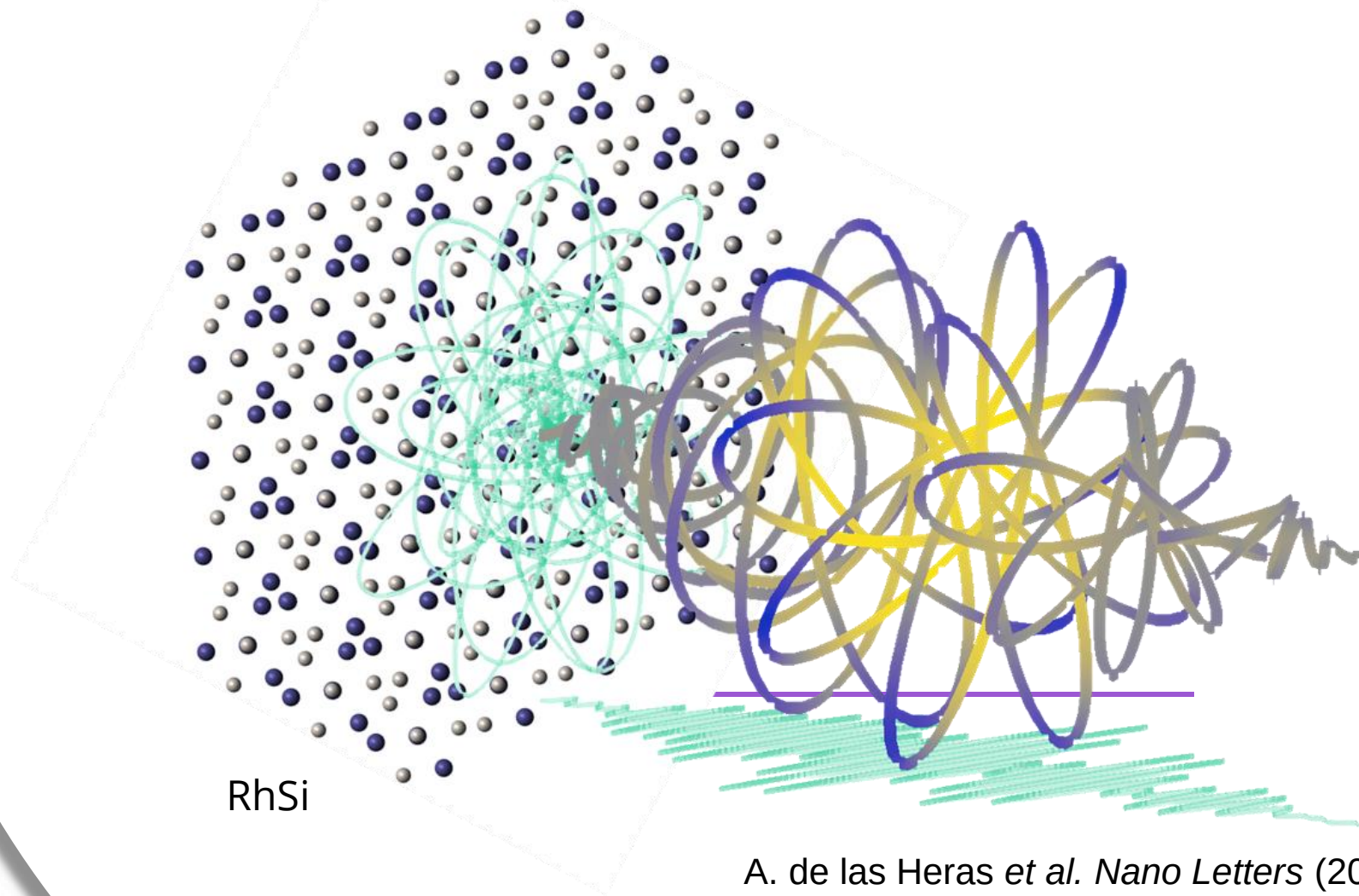
Role of pulse duration in solid-state high harmonic generation



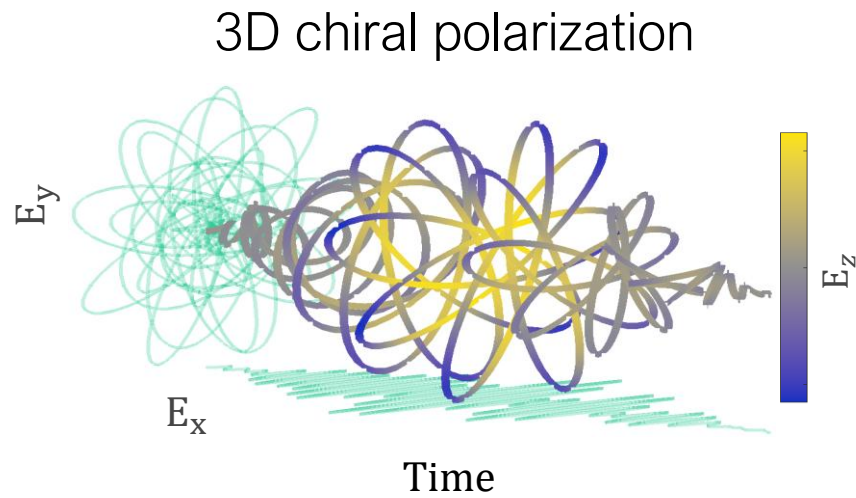
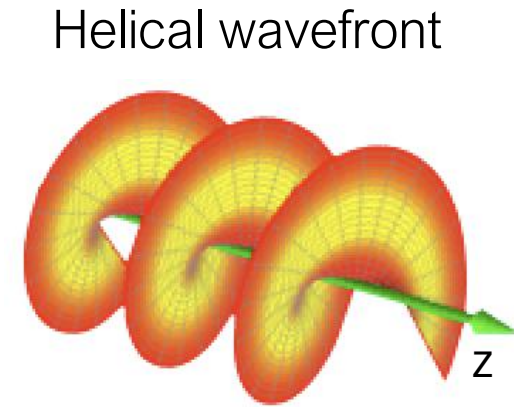
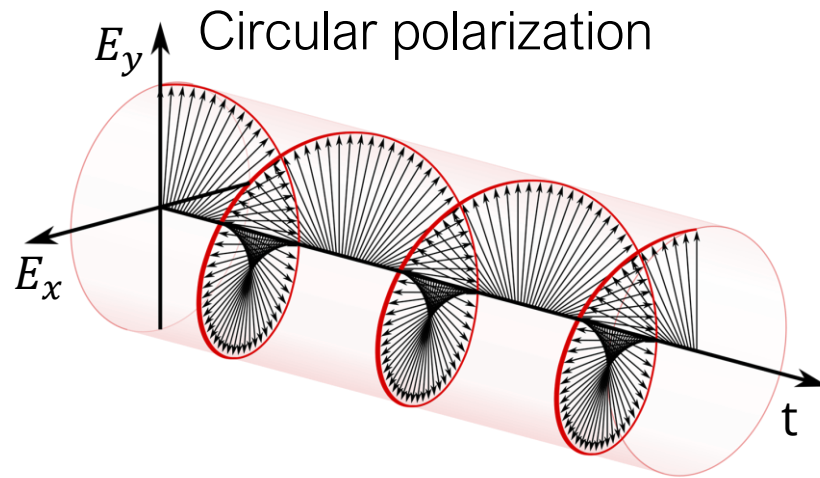
Pulse-duration sensitive high-harmonic extension survives the usual dephasing times



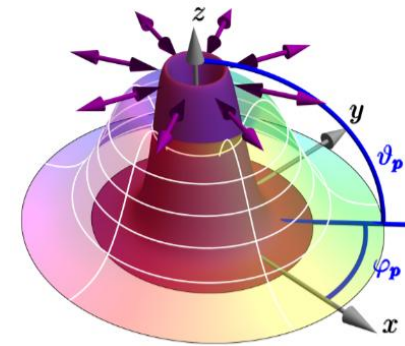
Generation of attosecond locally-chiral near fields



Overview of chiral light



Spatial polarization pattern

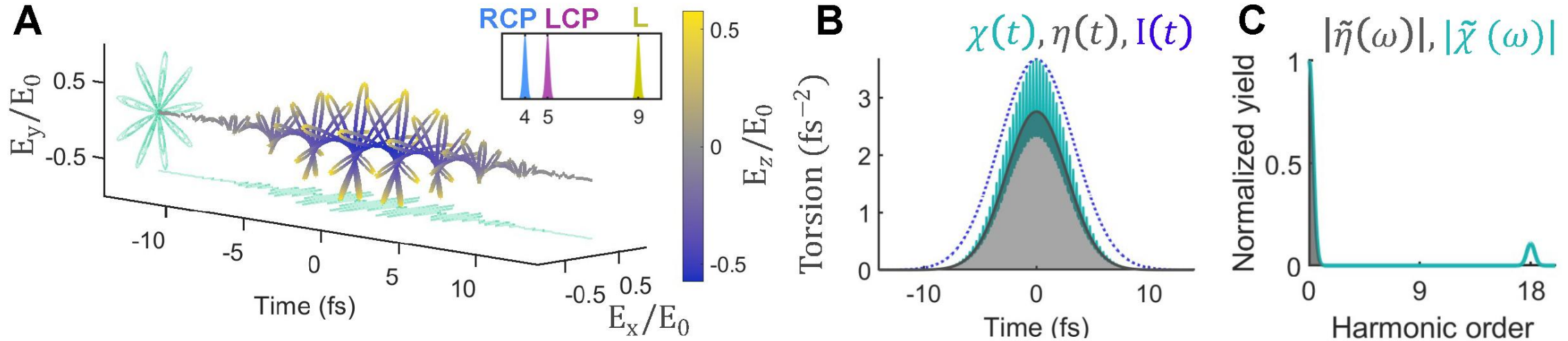


Locally-chiral light

3D chiral polarization

The electric field exhibits instantaneous torsion:

$$\chi(t) = \frac{\vec{E}(t)}{E_0^3} \cdot \left(\frac{d\vec{E}(t)}{dt} \times \frac{d^2\vec{E}(t)}{dt^2} \right)$$



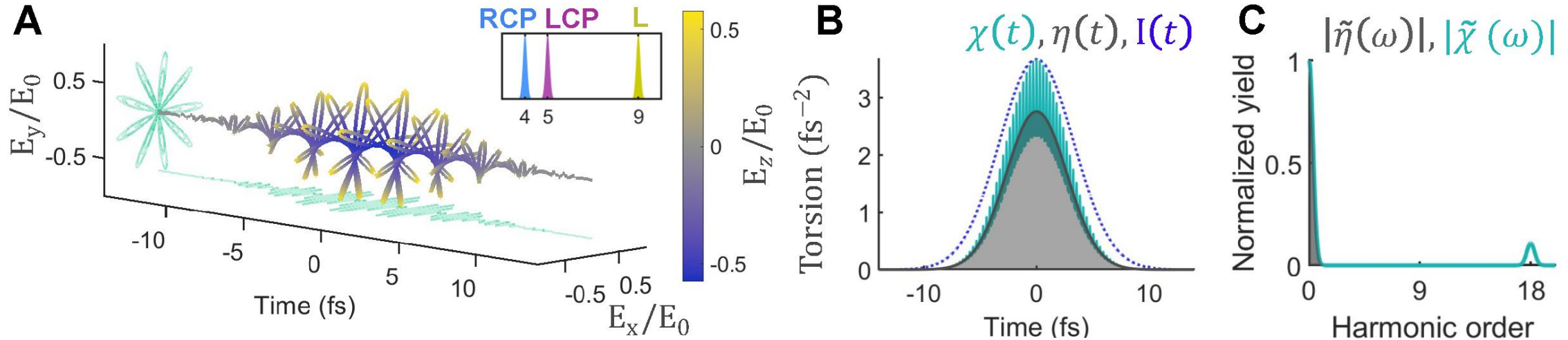
D. Ayuso *et al.* *Nat. Photonics* (2019)
A. de las Heras *et al.* [arXiv:2603.05346](https://arxiv.org/abs/2603.05346)

Locally-chiral light

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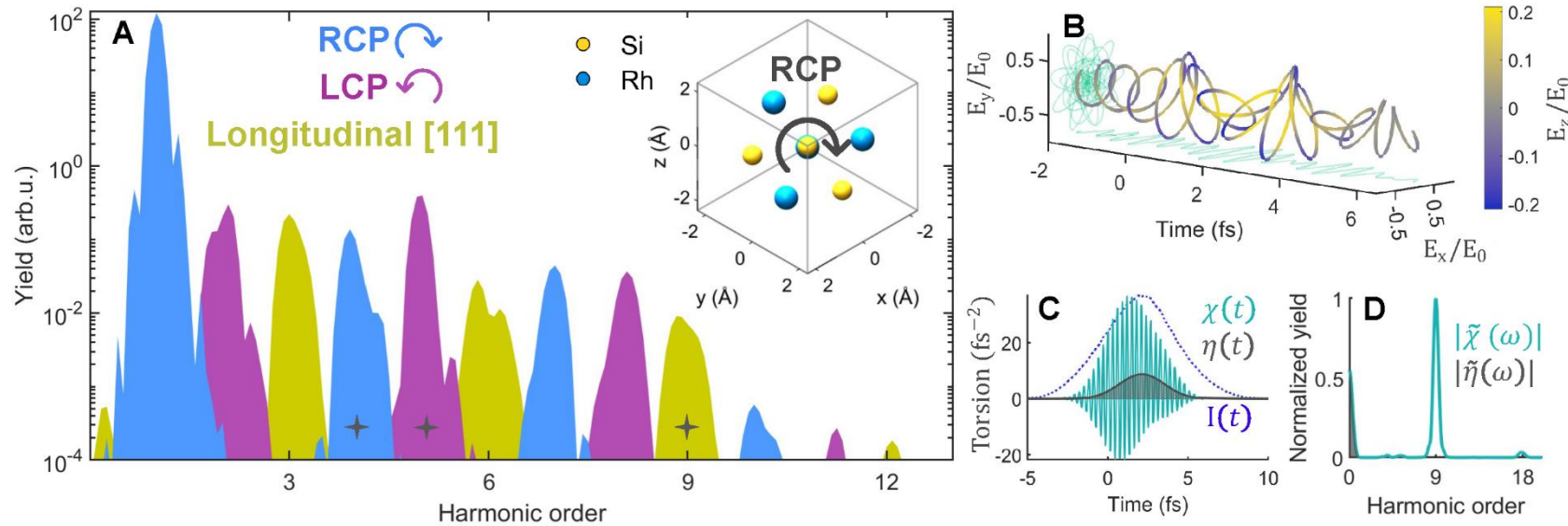
$$\chi(t) = \frac{\vec{E}(t)}{E_0^3} \cdot \left(\frac{d\vec{E}(t)}{dt} \times \frac{d^2\vec{E}(t)}{dt^2} \right)$$



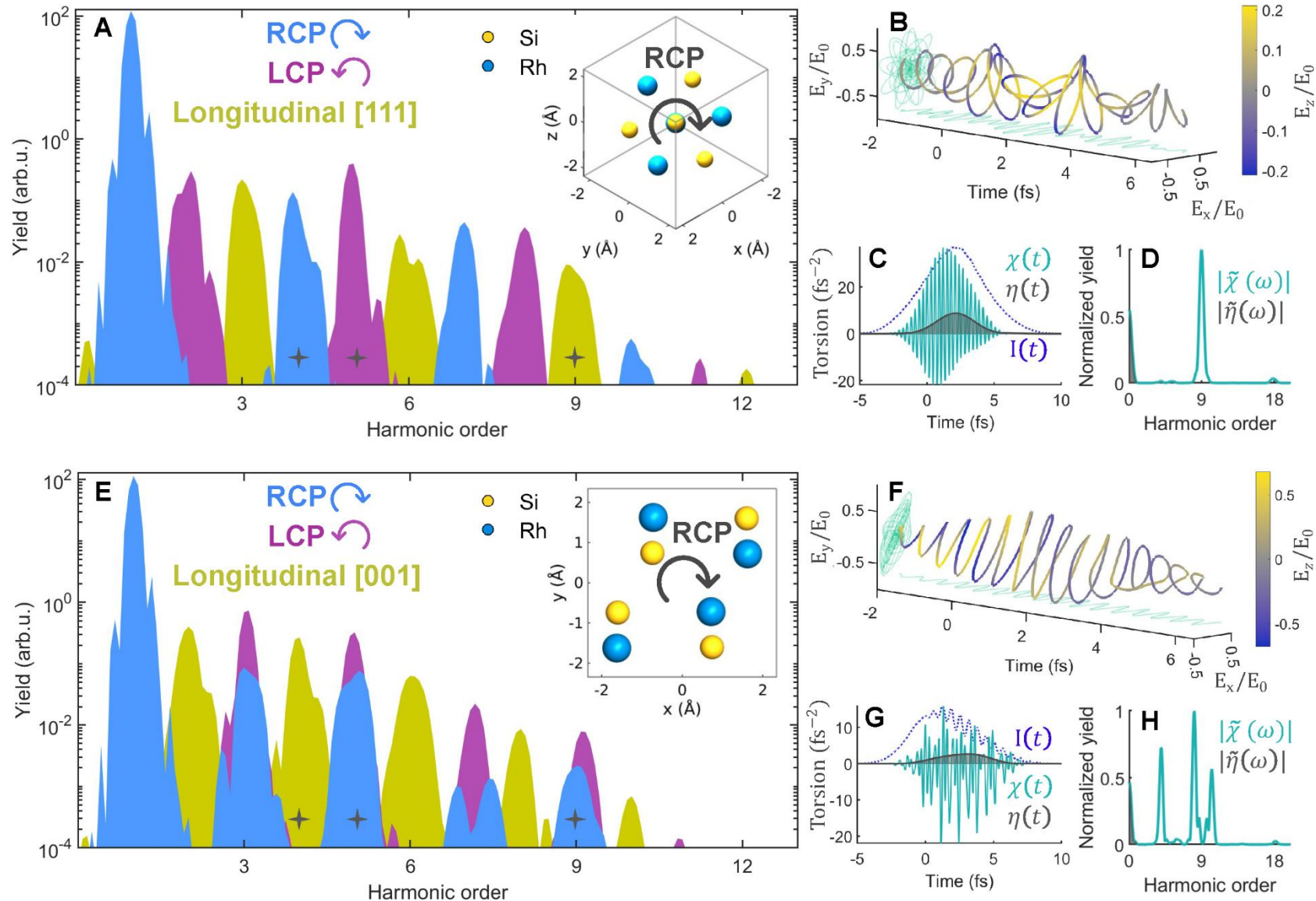
- The electric field vector is chiral at every spatial point
- Efficient chiral dipolar interaction, independently of the spatial scale
- The propagation of the z-polarization component is restricted in free space (transversality condition for plane waves $\vec{k} \cdot \vec{E} = 0$)
- Requires a multi-color synthesis
- Not measured in experiments yet

D. Ayuso *et al.* *Nat. Photonics* (2019)
A. de las Heras *et al.* [arXiv:2603.05346](https://arxiv.org/abs/2603.05346)

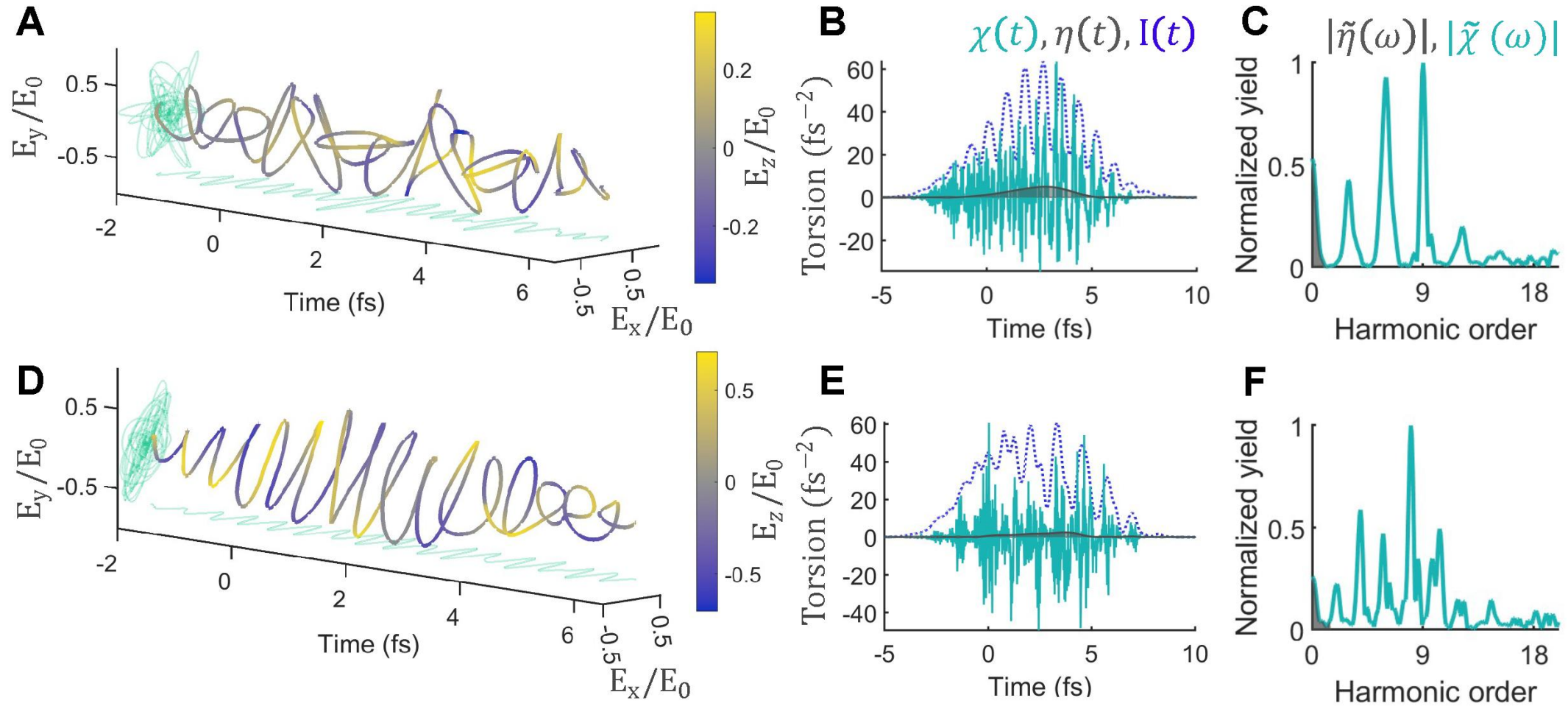
Synthesis of attosecond locally-chiral light in the interaction of RhSi with a right circularly polarized driver



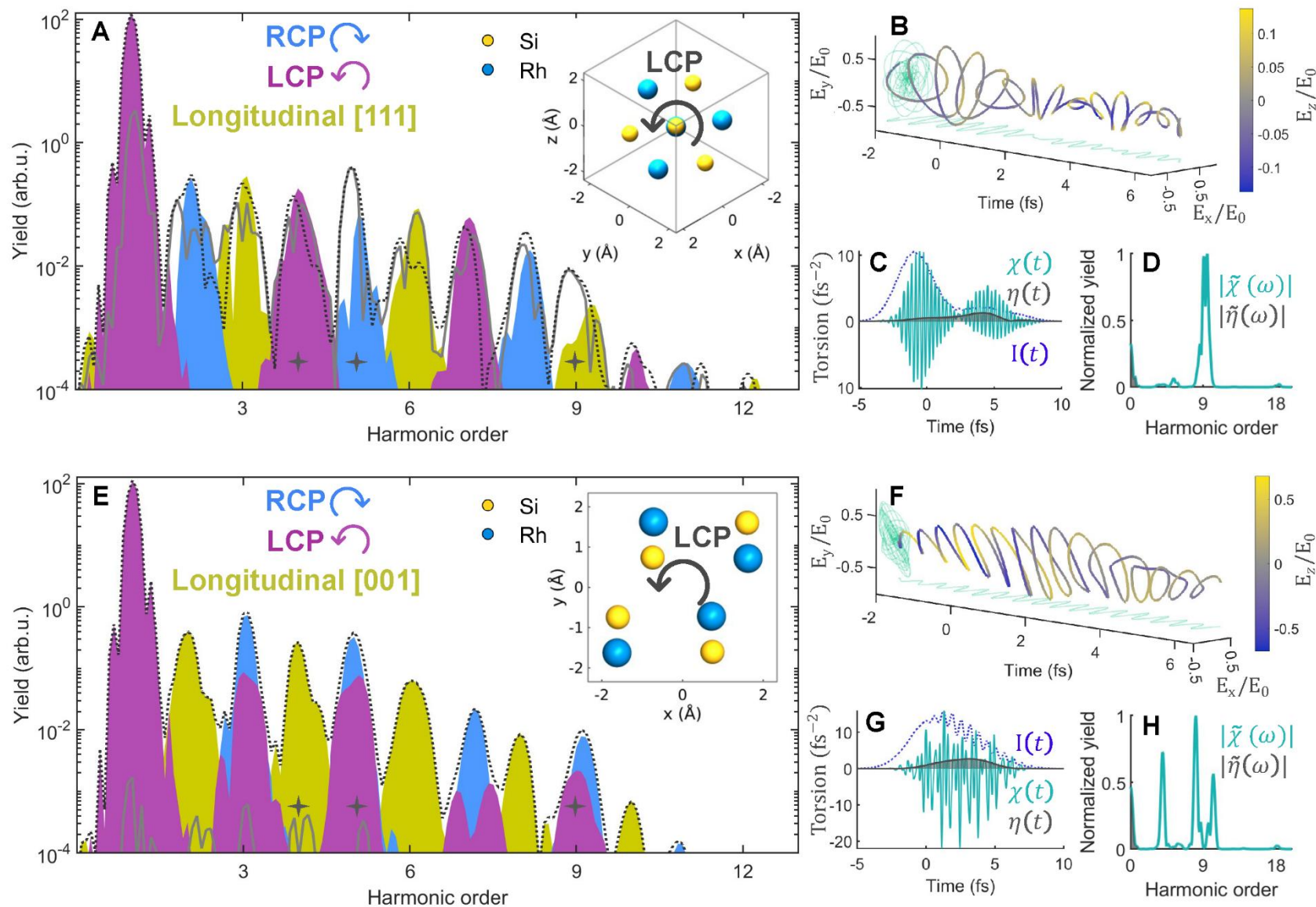
Synthesis of attosecond locally-chiral light in the interaction of RhSi with a right circularly polarized driver



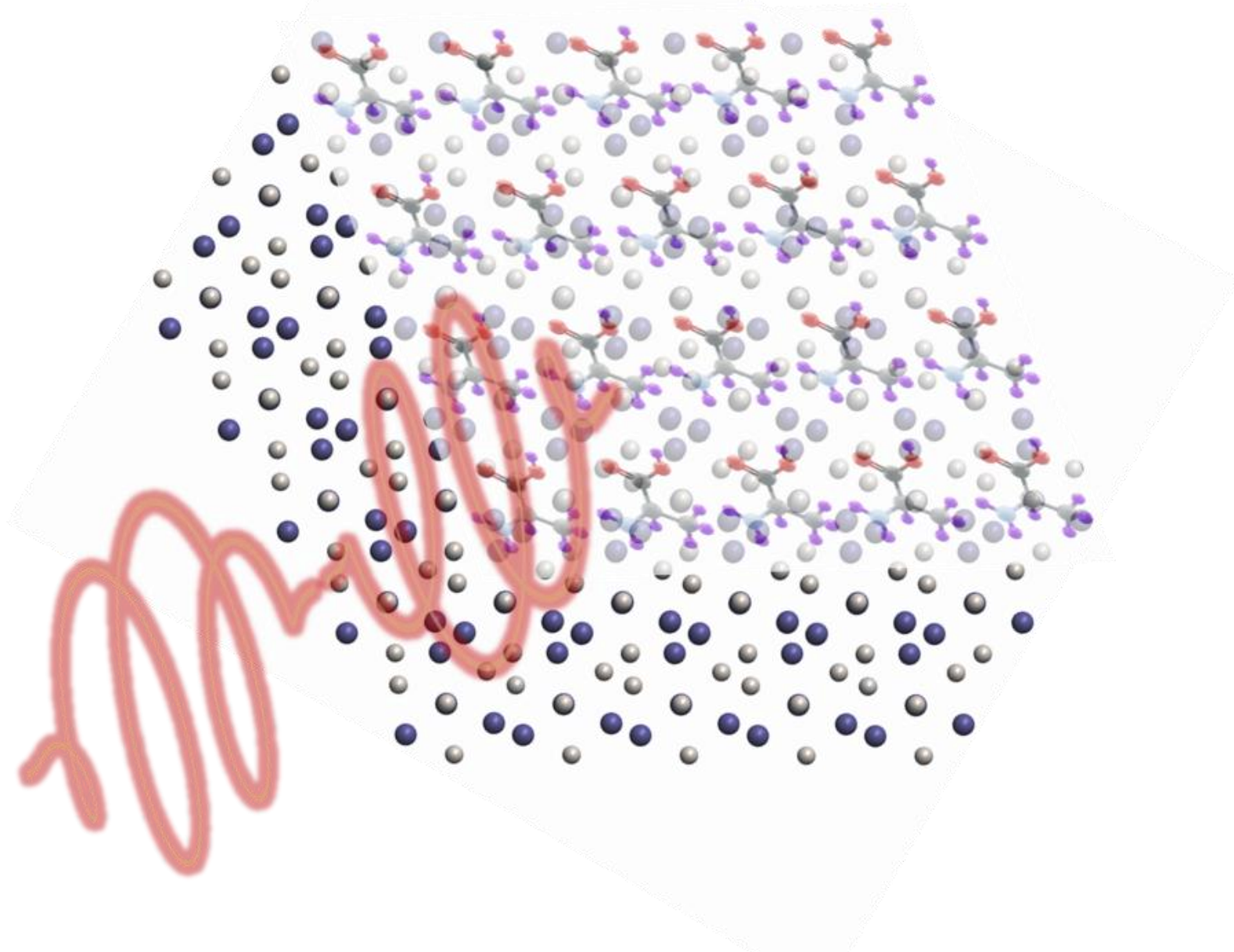
Attosecond locally-chiral light by filtering above a frequency threshold



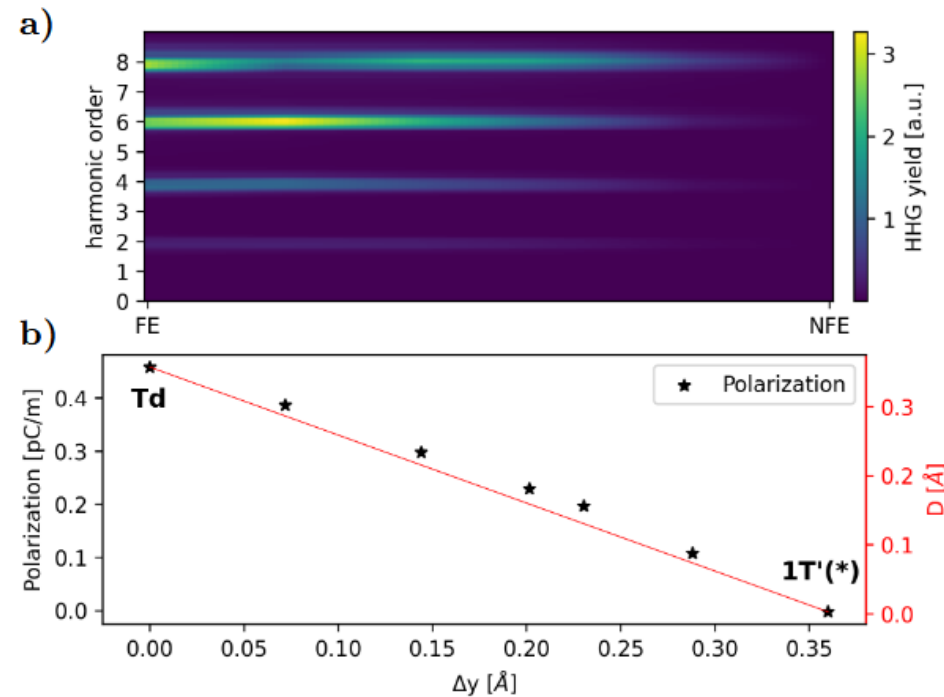
Circular dichroism in RhSi



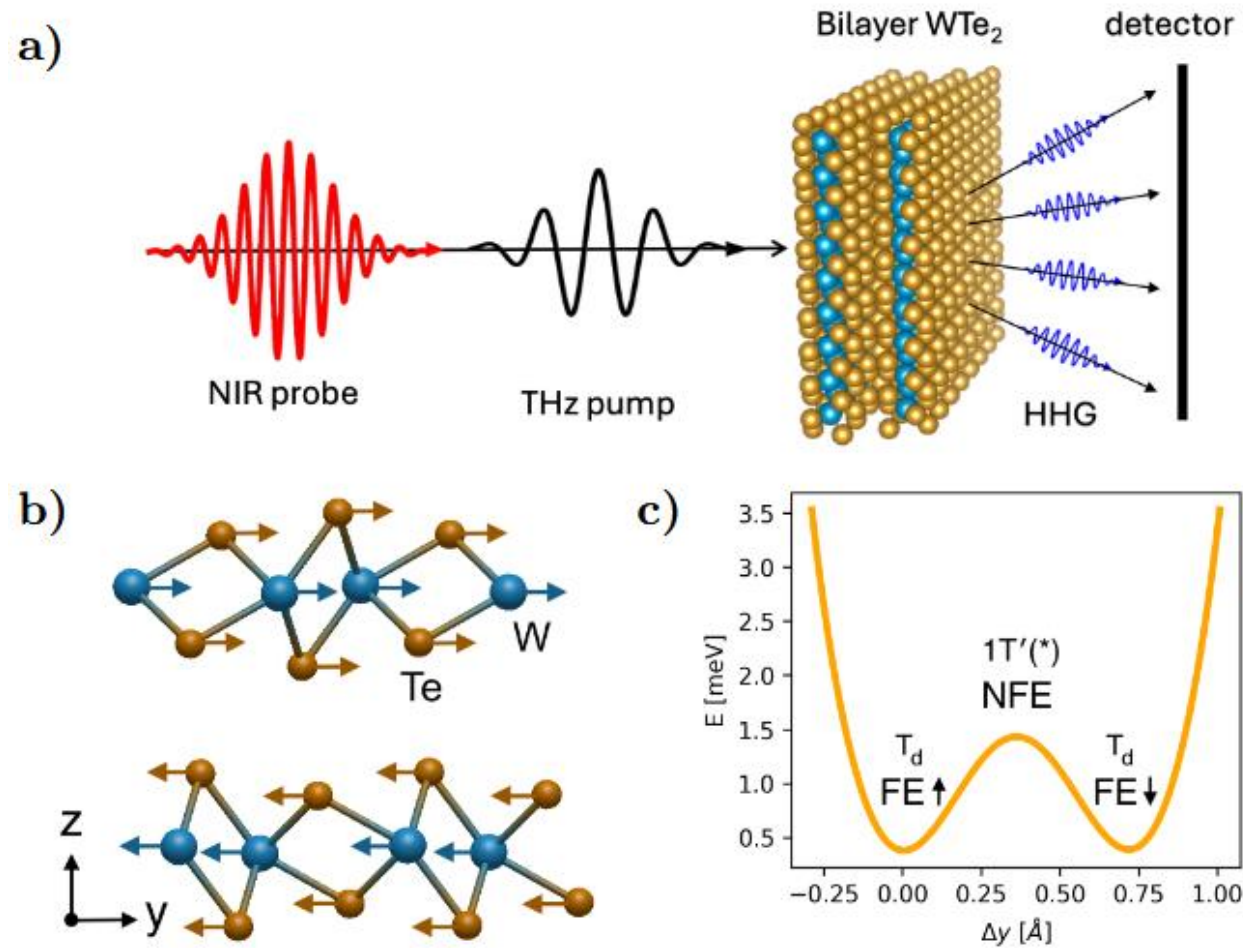
Application of attosecond locally-chiral light in a pump-probe scheme



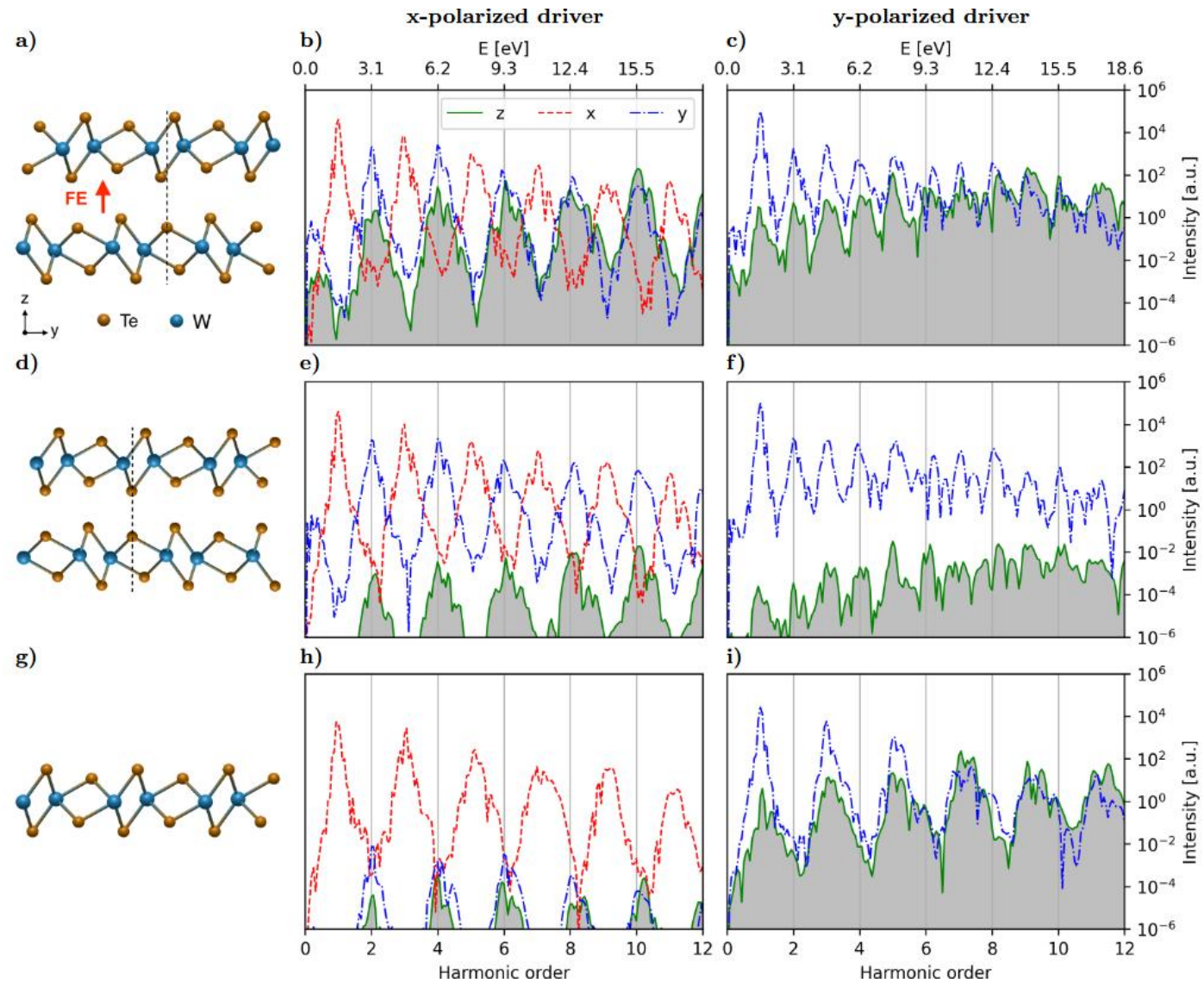
Probing sliding ferroelectricity in bilayer Td-WTe₂ with high-harmonic generation



THz pump – NIR probe setup for ferroelectric switching



High harmonic generation in Td-WTe₂



A matter of symmetries

Non-ferroelectric

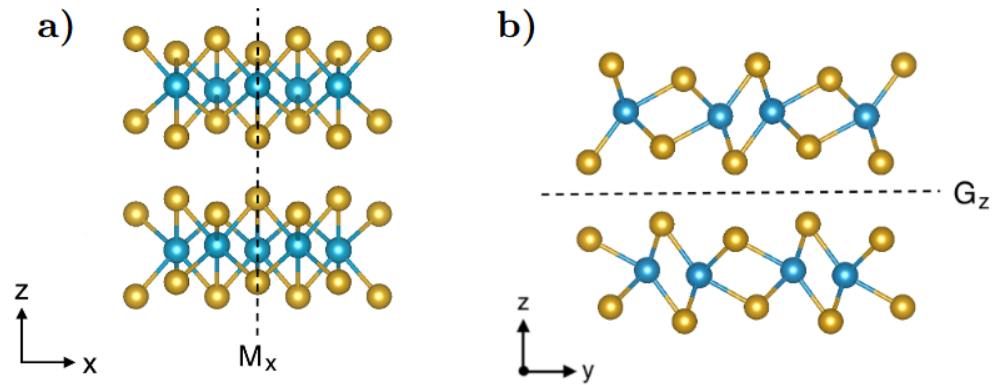


FIG. 5: **Symmetries of the non-ferroelectric $1T'(*)$ structure.** (a) Mirror plane M_x (corresponding to the yz plane). (b) Glide-plane symmetry G_z , consisting of a reflection across the xy plane combined with a fractional translation along the y direction.

Ferroelectric

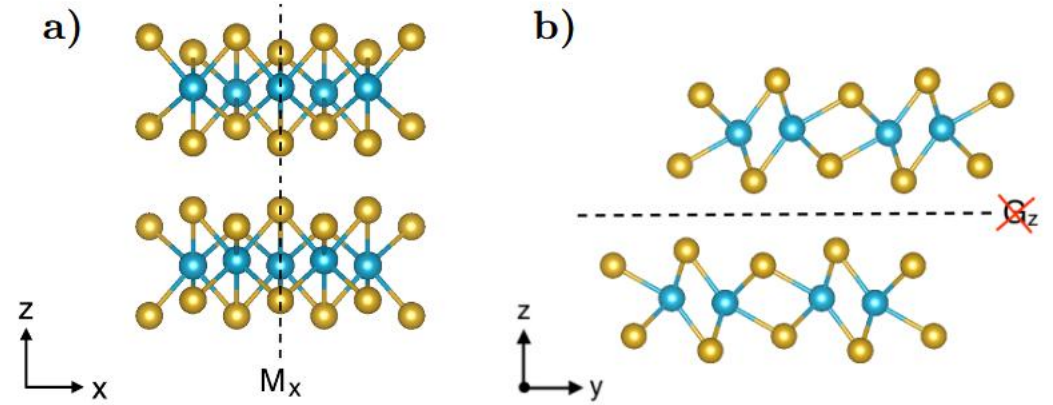


FIG. 6: **Symmetries of the ferroelectric T_d structure.** (a) Mirror plane M_x . (b) Illustration of the broken glide-plane symmetry G_z .

Using Octopus code for TDDFT calculations in solids



Octopus software



All the info in <https://octopus-code.org>

- Octopus is an **open-access** scientific software designed for **first-principles simulations of quantum systems**. It allows the computation of the electronic structure or dynamics using DFT or TDDFT. The Kohn Sham equations are solved in a real space grid for each time iteration.
- For time-dependent problems, we prepare 2 input files: one for the ground state calculation and another one to propagate it in time. We will name them “inp_gs” and “inp_td” respectively.
- Parameters such as the spatial and time steps, size of the numerical box, etc. need to be converged.
- For calculations of the band structure of a solid, we prepare 2 input files (“inp_gs” and “inp_bands”). In this case, we also tune the k-points grid for convergence.
- In the input files, we indicate the output quantities and choose the calculation options for a certain system.
- Some calculations require computer clusters, but simpler ones can run on a personal computer.

Exercises with Octopus software



E3. Generate the Octopus input files to calculate the **band structure** of **Silicon** considering a spatial step of $dx=0.4$ a.u. and a k-point grid of $16 \times 16 \times 16$. Check the crystal lattice parameters in <https://next-gen.materialsproject.org/materials>. Output the crystal geometry and the density of states in the ground-state calculation.

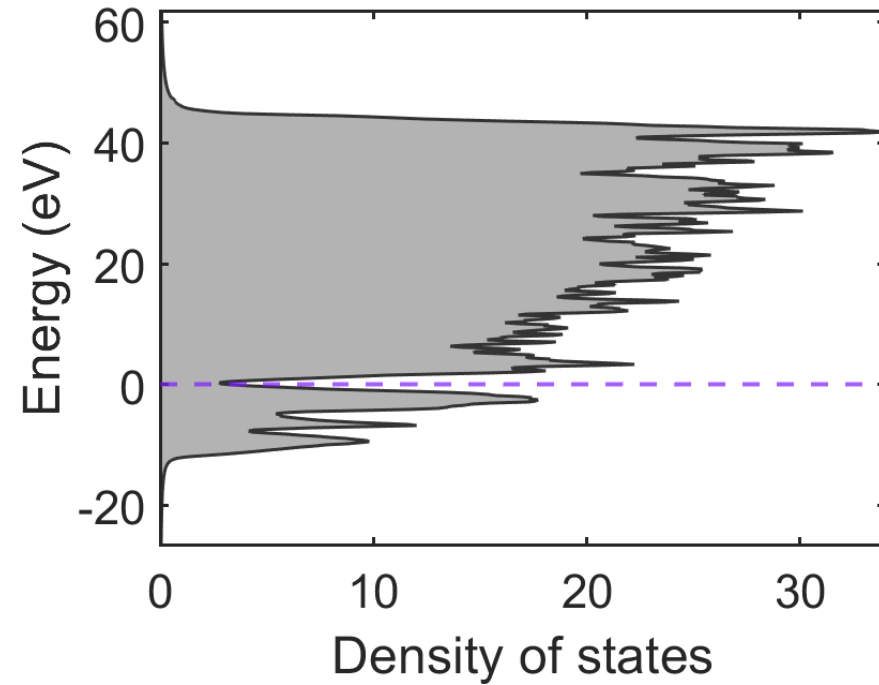
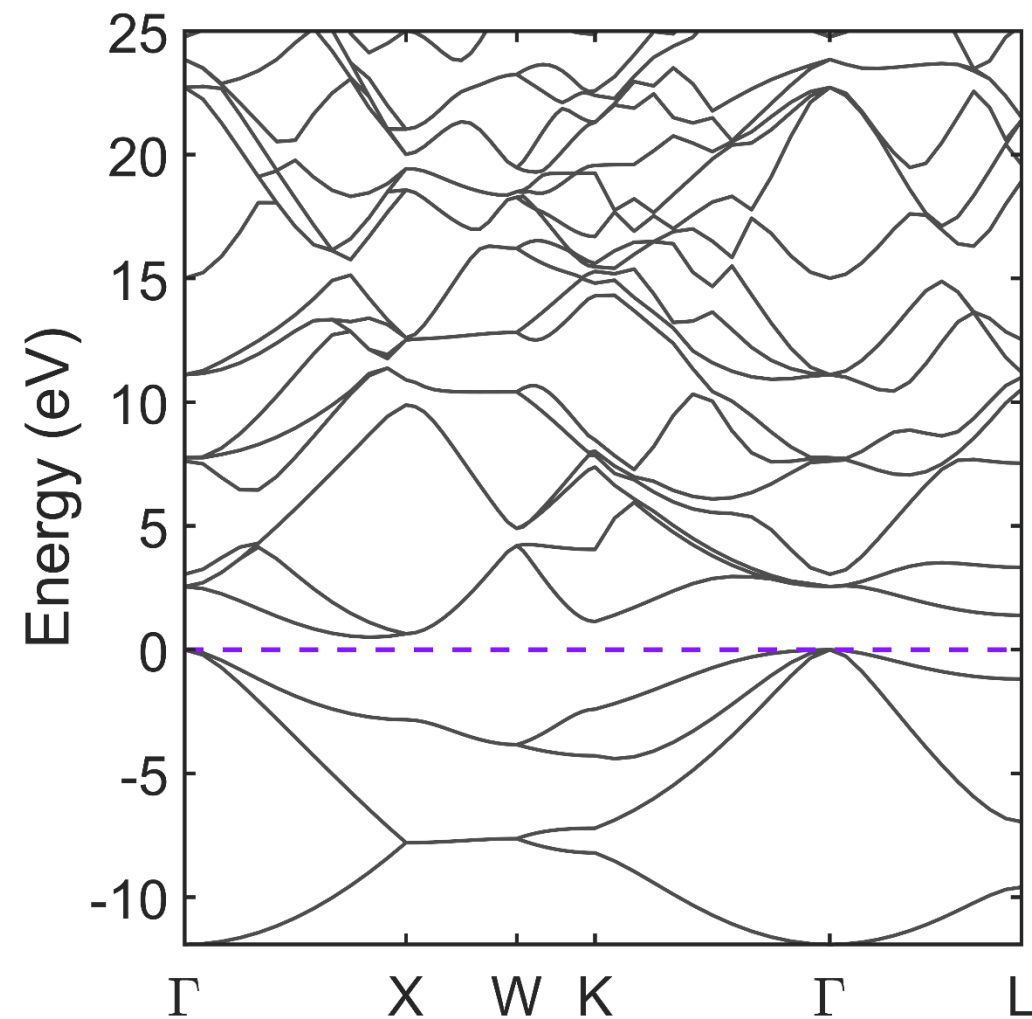
E4. Generate the Octopus input files to calculate the **high harmonic spectrum** of **Silicon** driven by a linearly polarized laser pulse of 12 cycles full duration, 1200 nm central wavelength, and 1×10^{12} W/cm² peak intensity along the [111] axis. For the time propagation, set a time step of $dt=0.3$ a.u.

Important files after executions with Octopus software



- General information about the execution in log file
- All the input variables are in `exec/parser.log`
- In ground state calculations, you can find information about the convergence in `static/convergence` and all the calculated ground state parameters in `static/info`. In solids, the crystal structure (geometry output) is stored in `static/geometry` and `static/crystal`
- After running an “unocc” calculation to compute the band structure, you will get the data in `static/bandstructure`
- In time-dependent calculations, the folder `td.general` contains the “TDOutput” data and `output_iter` the “Output” data at the specified instants of time

E3: Band structure of Silicon



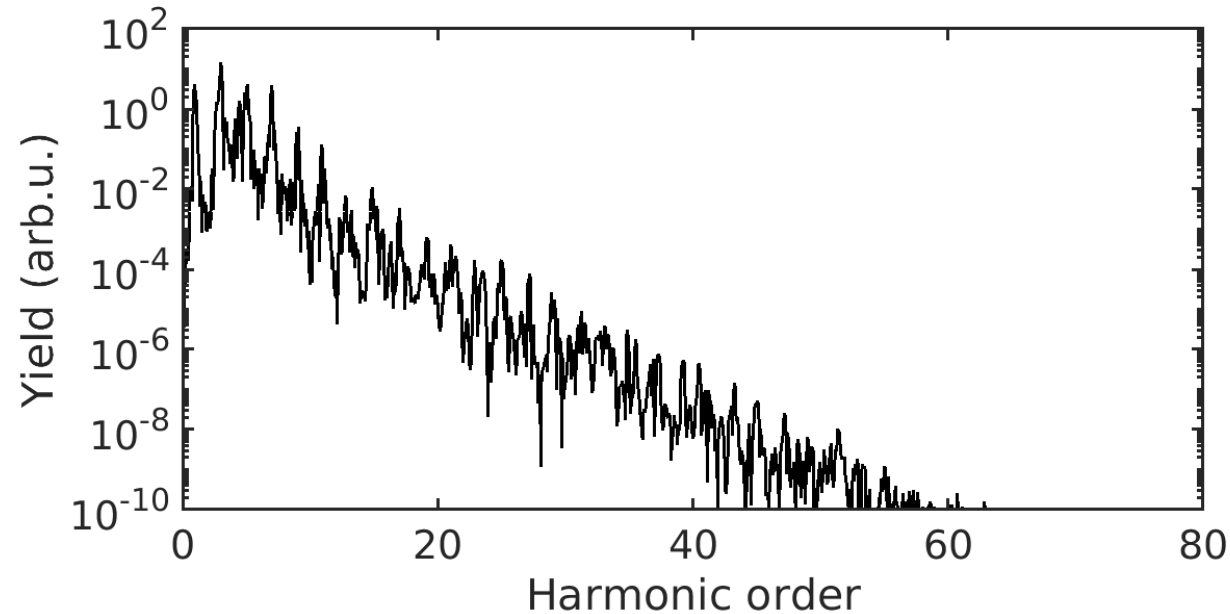
E4: High harmonic generation in Silicon

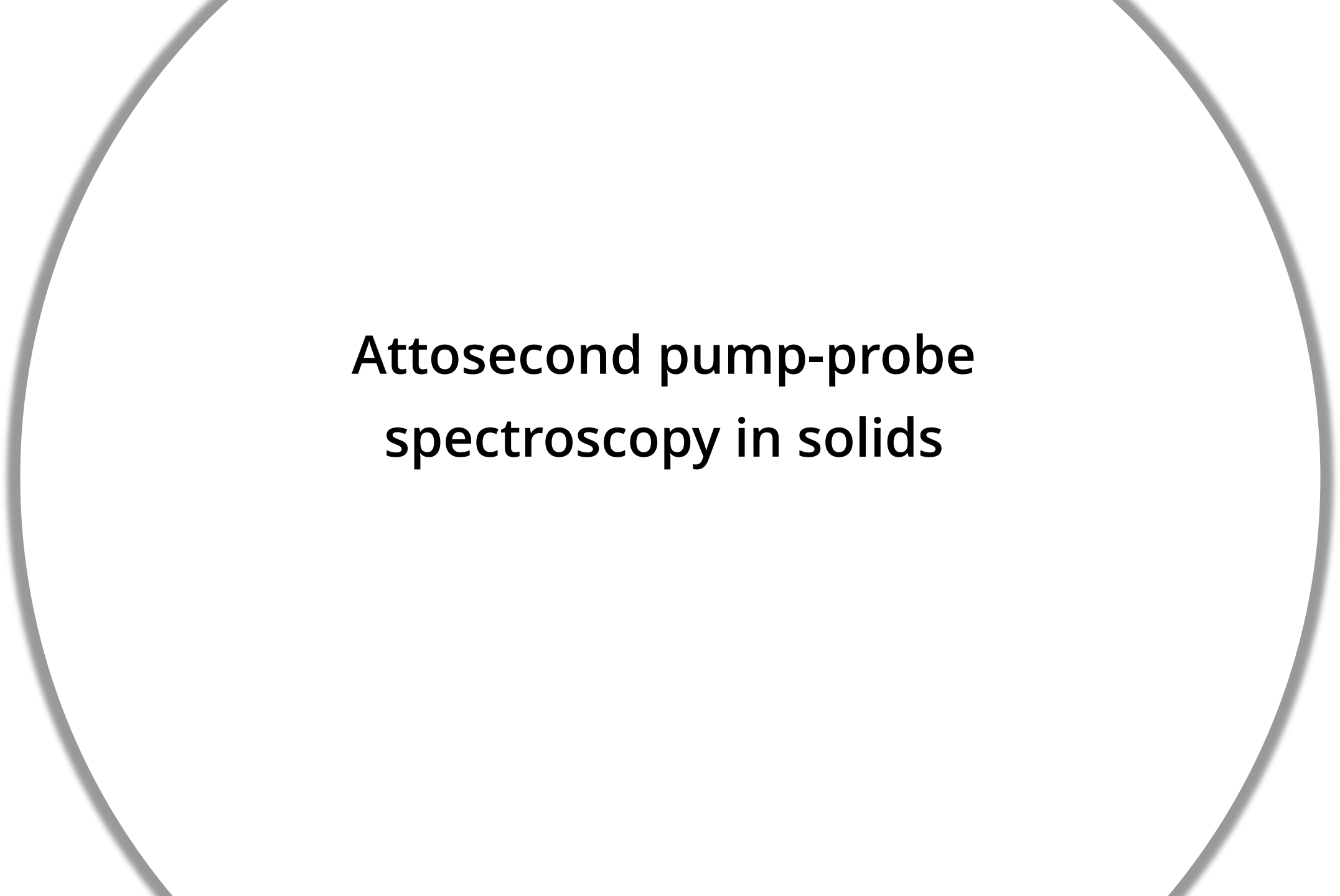
Driving vector potential:

$$\mathbf{A}(t) = A_0 \sin\left(\frac{\pi t}{\tau_{\text{pulse}}}\right)^{\frac{3\left|\frac{\pi(t)}{\tau_{\text{pulse}}} - \frac{\pi}{2}\right|}{4}} \cdot \Theta\left(\frac{\tau_{\text{pulse}}}{2} - \left|t - \frac{\tau_{\text{pulse}}}{2}\right|\right) \cos(\omega_0 t) \mathbf{u}_{\mathbf{p}} = A_{\text{env}}(t) A_0 \cos(\omega_0 t) \mathbf{u}_{\mathbf{p}}$$

Calculation of HHG spectra:

$$|\text{HHG}(\omega)|^2 = \sum_{\nu=x,y,z} \left| \int_{-\infty}^{\infty} \frac{dj_{\nu}(t)}{dt} e^{-i\omega t} A_{\text{env}}(t) dt \right|^2$$





Attosecond pump-probe spectroscopy in solids

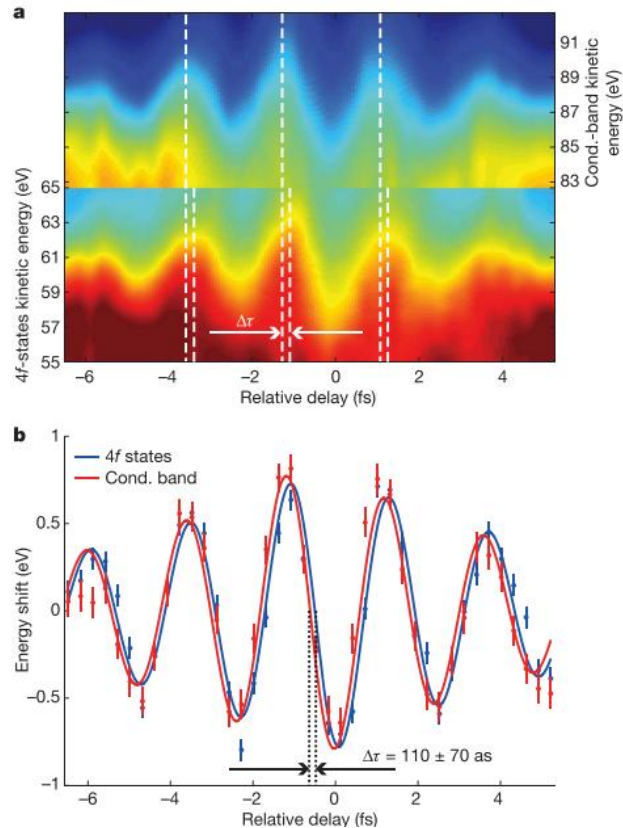
First experiment on attosecond spectroscopy in solids

Vol 449 | 25 October 2007 | doi:10.1038/nature06229

nature

Attosecond spectroscopy in condensed matter

A. L. Cavalieri¹, N. Müller², Th. Uphues^{1,2}, V. S. Yakovlev³, A. Baltuška^{1,4}, B. Horvath¹, B. Schmidt⁵, L. Blümel⁵, R. Holzwarth⁵, S. Hendel², M. Drescher⁶, U. Kleineberg³, P. M. Echenique⁷, R. Kienberger¹, F. Krausz^{1,3} & U. Heinzmann²



Attosecond streaking reveals a ~110 as delay between photoelectrons emitted from the conduction band versus the 4f localized core states in a tungsten crystal.

Simultaneous attosecond absorption and streaking spectroscopy

LETTER

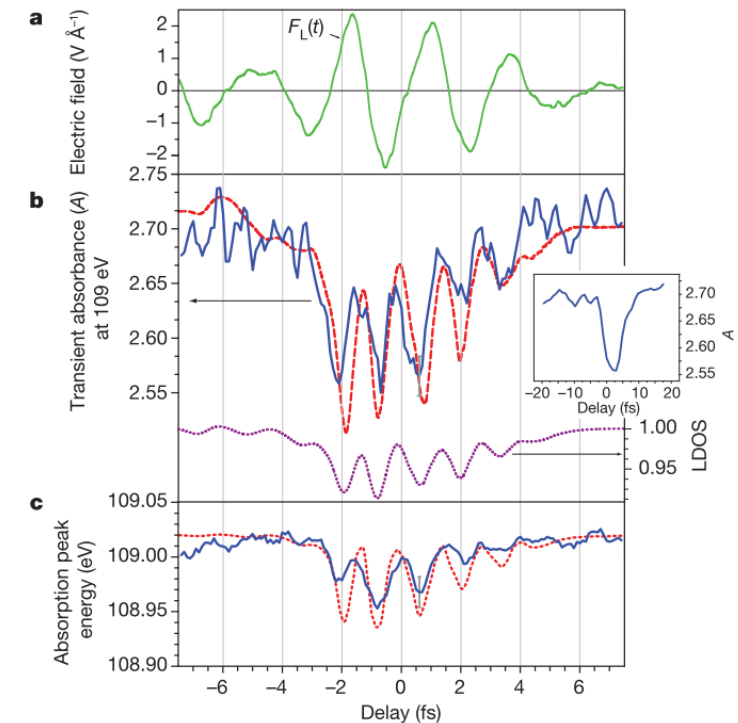
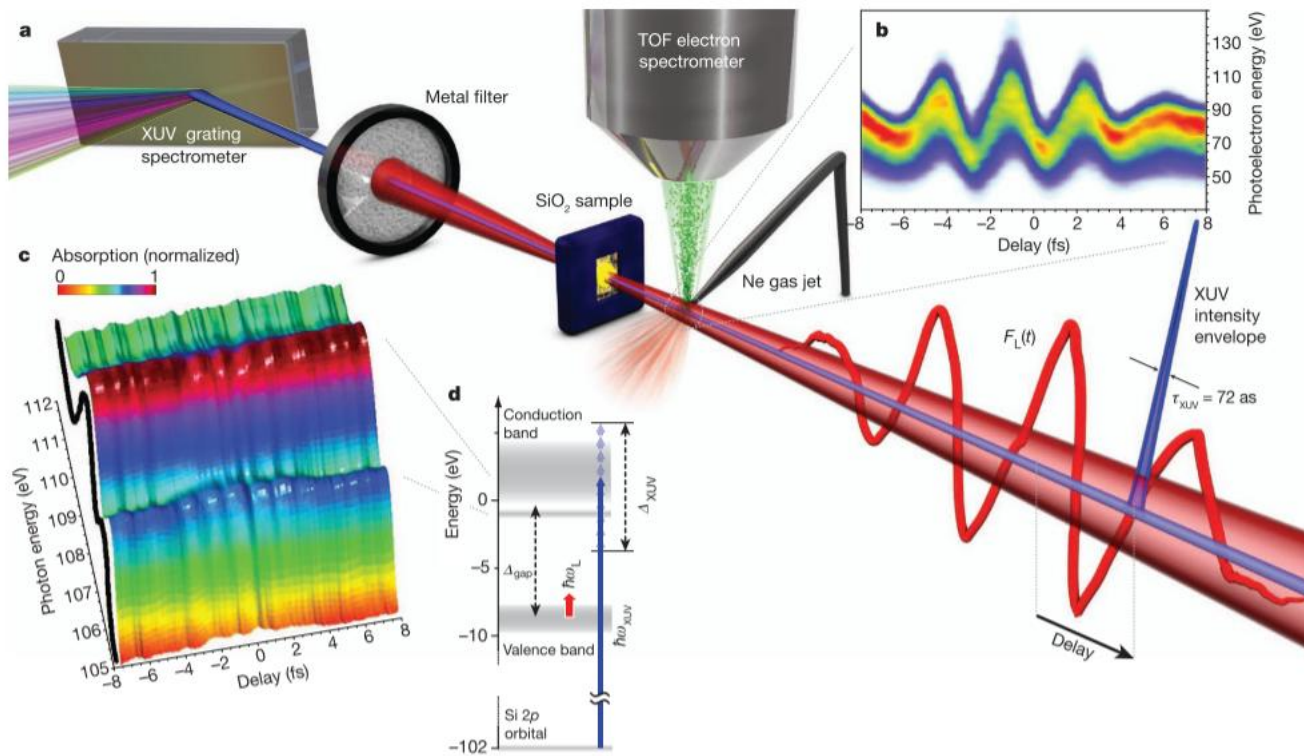
2013

doi:10.1038/nature11720

Controlling dielectrics with the electric field of light

Martin Schultze^{1,2}, Elisabeth M. Bothschafter^{1,3}, Annkatrin Sommer¹, Simon Holzner¹, Wolfgang Schweinberger¹, Markus Fiess¹, Michael Hofstetter², Reinhard Kienberger^{1,3}, Vadym Apalkov⁴, Vladislav S. Yakovlev², Mark I. Stockman⁴ & Ferenc Krausz^{1,2}

The absorption changes follow, in a highly nonlinear fashion, the turn-on and turn-off behavior of the driving field.



Attosecond many-body electron dynamics in transition metals

nature
physics

2019

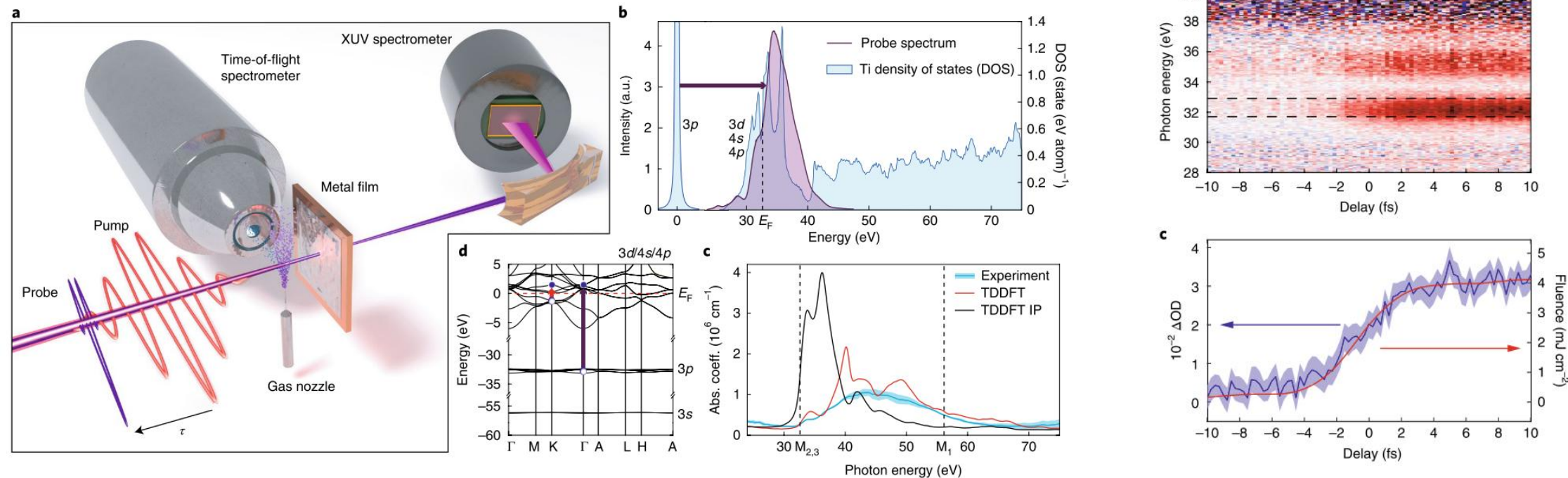
LETTERS

<https://doi.org/10.1038/s41567-019-0602-9>

Attosecond screening dynamics mediated by electron localization in transition metals

M. Volkov ^{1*}, S. A. Sato ^{2,4}, F. Schlaepfer ¹, L. Kasmi ¹, N. Hartmann ¹, M. Lucchini ^{1,5},
L. Gallmann ¹, A. Rubio ^{2,3} and U. Keller ¹

Quasi-instantaneous linear response in Ti (metallic system) due to ultrafast electronic localization on d-orbitals. The transient optical density follows the laser fluence.



Ultrafast core exciton dynamics in bulk MgF_2

ARTICLE Nature Communications 2021

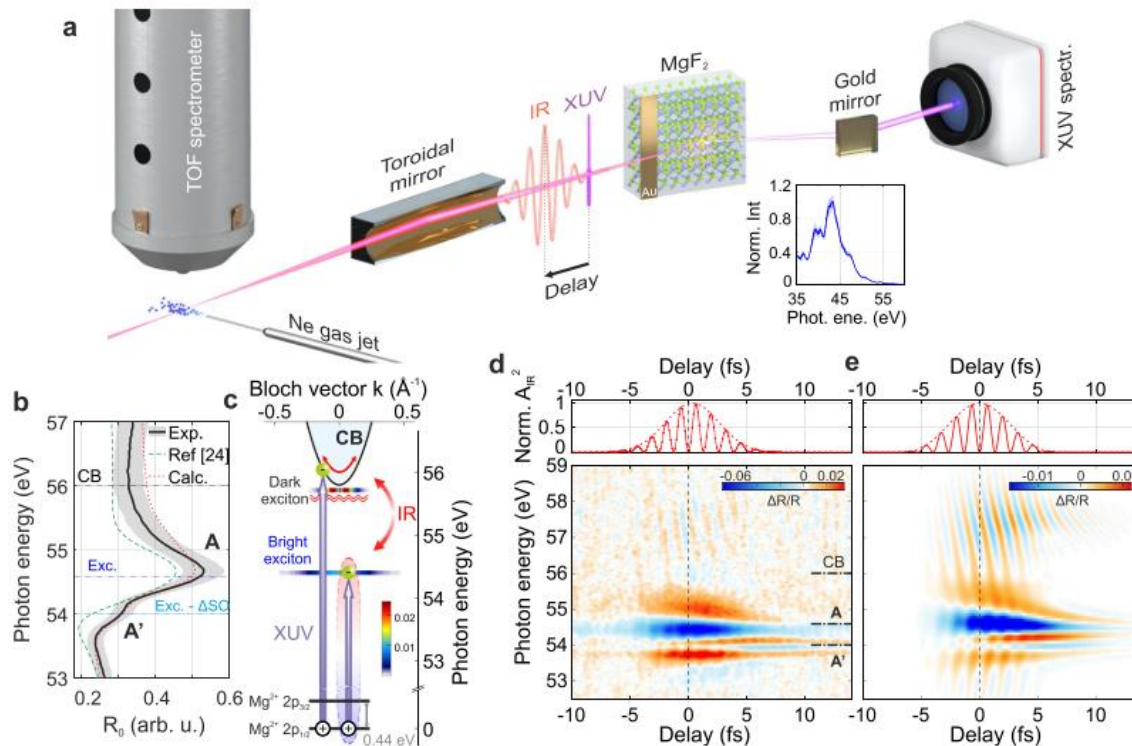


<https://doi.org/10.1038/s41467-021-21345-7>

OPEN

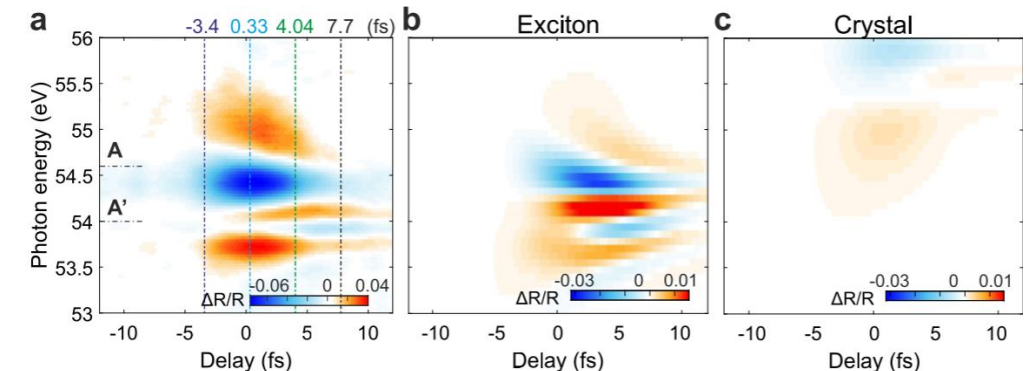
Unravelling the intertwined atomic and bulk nature of localised excitons by attosecond spectroscopy

Matteo Lucchini^{1,2}, Shunsuke A. Sato^{3,4}, Giacinto D. Lucarelli^{1,2}, Bruno Moio¹, Giacomo Inzani¹, Rocío Borrego-Varillas², Fabio Frassetto⁵, Luca Poletto⁵, Hannes Hübener⁴, Umberto De Giovannini^{4,6}, Angel Rubio^{4,6} & Mauro Nisoli^{1,2}



Disentangling core exciton dynamics with attosecond transient reflection spectroscopy:

- i) Optical Stark effect (atomic-like character, neglecting intraband motion)
- ii) Conduction band dynamical Franz-Keldysh effect (solid-like character)



Autoevaluation form of SESSION 4

<https://forms.gle/fvxHtdziMVbB79aR6>



Global feedback about the course

<https://forms.gle/vXi1AGf866vySdtp9>

